CausalLOB: Technical Foundation for a Dynamic Bayesian Network of Limit-Order-Book States

CSE 516 Probabilistic Graphical Models
Individual Project – Group G9

Milestone 2

Abstract

M1 defined CausalLOB as an observational probabilistic graphical model for short-horizon limit-order-book dynamics. M2 turns that proposal into an executable technical foundation. The FI-2010 NoAuction Decimal Precision data are reconstructed into six leakage-safe state variables and fifty boundary-separated day-by-security sequences. LPCMCI is used for lagged structure discovery on Days 1–5, candidate cross-variable families are challenged on Days 6–7, and the resulting graph is frozen before numerical parameter fitting. The M2 graph contains thirty own-history parent slots and nine validated cross-variable lag slots. After graph freeze, linear-Gaussian transition CPDs are fitted by Ridge regression and five ordered-probit outcome heads are fitted on Days 1–7. Exact Gaussian conditioning is then used for partial-observation inference. The first KPI results show modest graph-specific improvements over an equal-history AR(5) baseline in return reconstruction and selected missing-state queries, but not in every state variable or masking pattern. The result at M2 is therefore a working and reproducible PGM pipeline, not a claim of identified market causality or final predictive superiority.

1 M1 question and M2 scope

The original question is: *which lagged limit-order-book states are stable probabilistic antecedents of short-horizon midpoint-price formation, and can a Dynamic Bayesian Network (DBN) represent those dependencies for calibrated probabilistic inference?*

M1 fixed one domain, one principal dataset, a six-variable state representation and a maximum lag of five representation steps. It also separated the three PGM tasks: structure learning with LPCMCI, parameter learning after graph freeze, and posterior inference under partial observation. M2 now implements all three.

The causal language has been tightened during implementation. FI-2010 is observational. LPCMCI is therefore used to identify lagged temporal *candidates* under its assumptions. The project does not interpret an edge as an experimentally identified causal effect. A second clarification is that the fifty day-by-security blocks are boundary-separated sequences used to avoid temporal leakage; they are not treated as fifty independent experiments.

2 Data pipeline

The principal dataset is FI-2010 NoAuction Decimal Precision. The data-rebuild code reconstructs the canonical 394,337 representations over ten trading days and uses the raw order-book rows required by the original project rather than publisher-engineered explanatory features. Four large midpoint discontinuities within each day identify five anonymous security blocks. An independent reconstruction reproduces all fifty boundaries exactly.

The state vector is

$$X_t = \begin{bmatrix} s_t & d_t & q_t^{(1)} & q_t^{(2:5)} & o_t & r_t \end{bmatrix}^\top,$$

where s_t is log relative spread, d_t is log total L1–L5 depth, $q_t^{(1)}$ is L1 queue imbalance, $q_t^{(2:5)}$ is incremental L2–L5 imbalance, o_t is a depth-normalized quote-change OFI proxy and r_t is one-step log midpoint return. Rolling volatility remains a regime/sensitivity variable rather than a primary graph node.

No lag is allowed to cross a day/security boundary. With lags 1–5, the graph discovery and transition design matrices therefore use only rows with complete within-sequence history.

3 Graphical-model representation

For each target component j, the transition CPD is linear Gaussian:

$$X_{j,t} = c_j + \sum_{(i,\ell)\in \mathrm{Pa}_j} \beta_{ji\ell} X_{i,t-\ell} + \varepsilon_{j,t}, \qquad \varepsilon_{j,t} \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

The corresponding DBN factorization is

$$p(x_{1:T}) = p(x_{1:5}) \prod_{t=6}^{T} \prod_{j=1}^{6} p\big(x_{j,t} \mid x_{\mathrm{Pa}(j,t)}\big). \tag{2}$$

The primary frozen transition model uses a diagonal innovation covariance. This makes the conditional independence assumption explicit: after conditioning on the selected lagged parents, contemporaneous residual shocks are treated as independent in the primary DBN. Train-only correlated residual covariance is retained only as a sensitivity analysis, because changing Q changes the graphical assumptions.

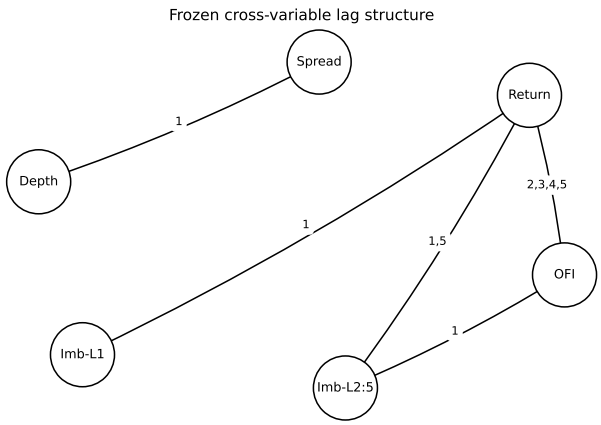


Figure 1: Frozen cross-variable structure. Edge labels are retained lag numbers; the thirty own-history lags are omitted from the drawing.

Every state variable keeps its five own-history lags, giving thirty fixed self-history parent slots. Structure discovery and D6–D7 validation add nine cross-variable slots from five source–target families. Figure 1 shows only the cross-variable part to keep the diagram readable.

The retained cross-variable families are:

- L1 imbalance → midpoint return, lag 1;
- L2–L5 imbalance → midpoint return, lags 1 and 5;
- L2–L5 imbalance → OFI proxy, lag 1;
- total depth → relative spread, lag 1;
- OFI proxy → midpoint return, lags 2–5.

Several orientations remain explicitly caveated because reciprocal feedback or ambiguous endpoints are plausible in market data.

4 Learning

4.1 Structure learning

LPCMCI is run only on Days 1–5 with maximum lag five. The algorithm is appropriate to the project because the series are strongly autocorrelated and latent common causes are plausible. The discovery stage produced 59 candidate links, including 27 directed cross-variable lag slots arranged into 16 directed source–target families.

Those families were not accepted automatically. On Days 6–7, each family was compared with an equal-history AR(5) Ridge model. The same validation also included a nonlinear predictor, an anti-causal lead placebo and an autocorrelation-preserving phase-shift reference. Five families survived the frozen predictive criteria. A post-discovery target-and-lag-matched random-set diagnostic gave a descriptive $p \approx 0.119$. This is one reason the graph is described as a validated sparse temporal representation rather than as established causal ground truth.

4.2 Parameter learning

After the graph is frozen, numerical parameters are fitted on Days 1–7. For each target, Ridge regression with $\alpha = 1$ estimates the coefficients in Eq. (1). The residual mean-square error supplies the diagonal innovation variance. The sparse DBN has 39 transition coefficients, six intercepts and six primary innovation variances.

Five ordered-probit heads are also fitted for the publisher horizons 10, 20, 30, 50 and 100 raw messages. These heads map the six-dimensional current state to three class probabilities. They are fitted only after the transition graph has been frozen.

5 Inference

The principal M2 inference query is the posterior distribution of missing current-state components given exact recent history and whatever current components are observed:

$$p(X_{t,m} \mid X_{t,o}, X_{t-1:t-5}).$$

Let the one-step DBN prior be

$$X_t \mid X_{t-1:t-5} \sim \mathcal{N}(\mu_t, Q),$$

and partition μ_t and Q into missing components m and observed components o. Exact Gaussian conditioning gives

$$\mu_{m|o} = \mu_m + Q_{mo} Q_{oo}^{-1} (x_o - \mu_o), \tag{3}$$

$$Q_{m|o} = Q_{mm} - Q_{mo} Q_{oo}^{-1} Q_{om}. \tag{4}$$

This is the primary inference algorithm because the model is linear Gaussian and the state dimension is only six; there is no reason to replace a closed-form calculation with an approximation for the main query.

The implementation nevertheless contains a Monte Carlo comparator for the M1 exact-versus-approximate commitment. It samples from the same conditional Gaussian and measures posterior-mean error versus runtime. In the deterministic six-dimensional Gaussian implementation test, posterior-mean RMSE falls from 0.228 with 32 samples to 0.0288 with 2,048 samples. This is a numerical inference check rather than market evidence, and the same comparator is configured to run on the frozen Sparse-39 DBN using Days 6–7. It changes neither the graph nor any fitted parameter.

For outcome inference, the conditional state posterior is propagated through the ordered-probit head analytically. The code checks covariance positive definiteness, probability normalization and exact/full/partial-mask edge cases before running the empirical queries.

6 Baselines and initial KPIs

The primary transition baseline is AR(5) Ridge with the same five lags, same training rows, same standardization

Table 1: Selected M2 KPIs. Positive "gain" values favor Sparse-39 over AR(5).

KPI	Value
Canonical observations	394,337
Boundary-separated sequences	50
Final DBN parent slots	39
Validated cross-variable families	5
AR(5) return RMSE	0.833930
Sparse-39 return RMSE	0.824720
Return RMSE gain	0.009211
State RMSE gain: miss return+OFI	0.004752
Joint NLL gain: miss return+OFI	0.013698
State RMSE gain: history only	0.002209
Outcome log-loss gain, $h = 10$	0.008034
Outcome log-loss gain, $h = 20$	0.005350

and same target variables as the sparse DBN. The comparison therefore changes the cross-variable structure rather than the amount of history.

Table 1 summarizes the first M2 numbers. They are preliminary technical-foundation KPIs, not final M3 statistical claims.

The transition result is deliberately modest. The sparse graph improves return reconstruction but leaves some targets unchanged because their additional validated parents do not enter those equations. Under exact missing-state inference, the graph helps most in the focal return-plus-OFI and history-only settings. It does not improve every mask: for example, the liquidity mask has a small negative state-RMSE gain. This mixed behavior is useful at M2 because it shows that the pipeline can measure when the graph helps and when it does not.

The ordered-probit heads also produce normalized three-class probabilities. On the historically exposed Days 8–10, the $h = 10$ sparse-DBN head has log loss 0.4994 versus 0.8303 for the training-prior baseline; at $h = 20$ the corresponding values are 0.7152 and 0.9586. These numbers demonstrate end-to-end operation but are kept separate from confirmatory evidence because those days were seen in earlier development.

7 SOTA anchor revisited

The M1 anchor is Gerhardus and Runge's LPCMCI work on high-recall causal discovery for autocorrelated time series with latent confounders. Its direct role in CausalLOB is *structure learning*, not state inference. The key transfer is the use of a method designed for strongly autocorrelated time series where hidden common causes are plausible.

Implementation changed the framing in two ways. First, LPCMCI output is not accepted as the final graph by itself; cross-variable families are subjected to out-of-discovery predictive falsification before freeze. Second, the final object is an explicit parametric DBN whose CPDs and inference queries can be inspected independently of the discovery algorithm. This separation between discovery, parameter estimation and inference is now clearer than it was at M1.

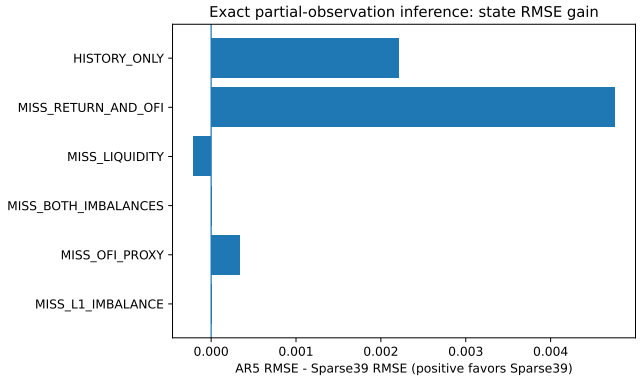


Figure 2: Initial exact-inference state-RMSE gain over the equal-history AR(5) baseline. Not every masking pattern benefits from the sparse graph.

8 Limitations and M3 handoff

The present milestone establishes machinery, not final evidence. FI-2010 remains observational, securities are anonymous, and the publisher labels cannot be reverse-engineered exactly from the ten-event representation rows alone. The fifty sequences prevent boundary leakage but do not create fifty independent experimental units. Most importantly, Days 8–10 were historically inspected and are therefore retrospective.

M3 should keep the M2 graph frozen while expanding statistical analysis: repeated block-level uncertainty, stronger ablations, calibration analysis, inference error–runtime curves on the real frozen DBN, and sensitivity to the innovation-covariance assumption. The central M2 deliverable is now in place: representation, learning, parameter estimation, inference, baseline comparison and measurable KPIs all execute end to end.

References

[1] A. Ntakaris et al. FI-2010 limit-order-book benchmark dataset and mid-price forecasting setup. arXiv:1705.03233.

[2] A. Gerhardus and J. Runge. "High-Recall Causal Discovery for Autocorrelated Time Series with Latent Confounders." *Advances in Neural Information Processing Systems*, 2020.